

PRACTICAL – 1:

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nirvighna@DESKTOP-VS57D2  ×  +  ∨  -  □  ×

GNU nano 7.2                                practicall.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

#define MAX_INPUT 200
#define MAX_ARGS 20

int main()
{
    char input[MAX_INPUT];
    char *args[MAX_ARGS];
    int arg_count = 0;

    printf("Linux Command Execution using fork(), exec() and wait()\n");
    printf("Enter a Linux command: ");

    if (fgets(input, sizeof(input), stdin) == NULL)
    {
        printf("Error reading input.\n");
        return 1;
    }

    input[strcspn(input, "\n")] = '\0';

    /* Split input into command and arguments */
    char *token = strtok(input, " ");

    while (token != NULL && arg_count < MAX_ARGS - 1)
    {
        args[arg_count++] = token;
        token = strtok(NULL, " ");
    }

    args[arg_count] = NULL;

    ^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
    ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify

nirvighna@DESKTOP-VS57D2I:~/OS-Week1/src$ ls -l
total 20
-rw-r--r-- 1 nirvighna nirvighna 6065 Sep 21 04:44 history_shell.c
-rw-r--r-- 1 nirvighna nirvighna 1783 Sep 21 04:04 mini_shell.c
-rw-r--r-- 1 nirvighna nirvighna 3715 Sep 21 05:06 parser.c
-rw-r--r-- 1 nirvighna nirvighna 1702 Sep 21 05:29 practicall.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1/src$
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nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ la -l bin
total 20
-rwxr-xr-x 1 nirvighna nirvighna 19784 Sep 21 05:34 practical1
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
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nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ la -l bin
total 20
-rwxr-xr-x 1 nirvighna nirvighna 19784 Sep 21 05:34 practical1
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/practical1
Linux Command Execution using fork(), exec() and wait()
Enter a Linux command: ls

Parent Process
Parent PID: 4257
Child PID: 4260
Parent waiting for child...

Child Process
Child PID: 4260
Parent PID: 4257
Executing command...

Makefile bin mini_shell.c src
Child process completed.
Parent process completed.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
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